

Facilitation for non-identical cognates in L3

The role of cross-language similarity and stimulus list composition

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In the present study we aimed to obtain with trilinguals the cognate inhibitory effects reported earlier for bilingual speakers who performed L2 lexical decision tasks (LDTs) with non-identical cognates and controls. To that end, Polish–English–Dutch trilinguals performed two LDTs in their L3. In Experiment 1, the stimuli included two types of double non-identical cognates Polish–Dutch (e.g., *SMAK–SMAAK*), and English–Dutch (e.g., *BUTTER–BOTER*) as well as matched non-cognate Dutch controls (e.g., *JASJE*). In Experiment 2, we tested triple non-identical cognates shared across Polish, English and Dutch (e.g., *GRUPA–GROUP–GROEP*) and Dutch controls (e.g., *BROEK*). We failed to find the bilingual inhibitory effects. In contrast, significant facilitation for English–Dutch and for Polish–English–Dutch cognates was found, even though no identical cognates were used. However, Polish–Dutch cognates yielded null results. The current findings emphasize the influence of the levels of form similarity on the observed cognate effects and point to the important role of stimulus list composition in cognate processing.

Keywords: non-identical cognate, trilingual, word recognition, stimulus list, cross-language similarity

1. Introduction

The defining feature of cognates is cross-language overlap in form and in meaning (e.g., *PARK* denotes a green area with trees both in Polish and in English). Cognates are frequently used during research testing whether all languages known to a speaker are co-activated, even when only one of those languages is relevant for the task at hand (Dijkstra et al., 2010; Lijewska, 2020). Generally, cognates, relative to control words (non-cognates), yield a processing advantage (i.e., faster and

more accurate responses) across a variety of tasks and paradigms. This advantage in cognate processing has been termed *cognate facilitation effect* and it is typically taken to reflect an integrated lexicon and the co-activation of the native language (L1) and the second language (L2) during lexical access (Dijkstra et al., 2010; Dijkstra & van Heuven, 2002; Lemhöfer et al., 2004; Martínez García, 2020; Szubko-Sitarek, 2011, 2015; Voga & Grainger, 2007).

The form overlap across languages can be complete, as in *identical cognates* (e.g., *FILM*–*FILM* across Polish and English) or incomplete, as in *non-identical cognates* (e.g., *TUNEL*–*TUNNEL* across Polish and English) (Dijkstra et al., 2010; Vanlangendonck et al., 2020). Furthermore, once speakers of more than two languages are considered (e.g., trilinguals), cognates can be shared across two languages and then they are referred to as *double cognates* (e.g., *ZIMA*–*WINTER*–*WINTER* or *SMAK*–*TASTE*–*SMAAK* for Polish–English–Dutch trilinguals) or across all three languages (e.g., *ECHO*–*ECHO*–*ECHO*) and then they are called *triple cognates*. These cognates can also be identical or non-identical, depending on the degree of form overlap of the words across the languages in question.

Identical cognates typically induce facilitation during processing both in bilingual and in trilingual speakers (Dijkstra et al., 2010, 2023; Van Assche et al., 2011; Szubko-Sitarek, 2015; Lemhöfer et al., 2004). The cognate advantage has been differently accommodated by various models of lexical processing in bilinguals (see Dijkstra et al. 2010, for an overview). Within the Bilingual Interactive Activation Plus model (BIA+; Dijkstra & van Heuven, 2002), and within its implemented version Multilink (Dijkstra et al., 2019), cognate facilitation is conceived as orthographic-semantic priming. Namely, upon the presentation of a cognate, similar form representations in two languages are co-activated and jointly activate the semantic representation thereby leading to speeded and more accurate responses to cognates relative to non-cognate controls (i.e., cognate facilitation). Within BIA+ and Multilink, non-identical cognates, whose cross-language overlap is smaller than that in identical ones, can benefit from cross-language co-activation only to a limited extent, hence they produce a significantly smaller cognate advantage. Even though BIA+ and Multilink were originally proposed as models of visual word processing in bilingual speakers, similar processing principles have been proposed by Dijkstra (2003) for trilinguals (see also Aboh & Parafta Couto, 2024).

Interestingly enough, prior bilingual research has shown that if only non-identical cognates are tested in a task (i.e., without any identical ones in the stimulus list), the cognate facilitation effect may vanish (Duyck et al., 2007) or even turn into cognate inhibition (i.e., slower and less accurate responses to cognates relative to controls as reported by Comesaña et al. (2012, 2015)). To the best of

our knowledge, the current study is the first to investigate the processing of non-identical double and triple cognates only (without identical cognates in the stimulus list) in a lexical decision task (LDT; where participants decide whether a presented string of letters is a word in a given language or not) in trilinguals' third language (L3). Thereby, we aimed to test whether findings of inhibitory effects of non-identical cognates reported earlier by Comesaña et al. (2012, 2015) for bilinguals can be found when non-identical cognates are tested with trilingual participants. This would provide further evidence for lateral inhibition operating in LDT as a function of specific stimulus list composition postulated by Comesaña et al. (2015; but see also Dijkstra et al., 2023).

However, it needs to be noted that there is only a small number of experiments which have tested cognate processing in trilinguals and many of those studies showed processing patterns unattainable with bilinguals (Lemhöfer et al., 2004; Lijewska & Błaszowska, 2021; Lijewska & Chmiel, 2015; Poarch & Van Hell, 2012, 2014; Szubko-Sitarek, 2011, 2015; Zhu & Mok, 2020). The present study aims to contribute to this research by investigating the processing of double and triple non-identical cognates when no identical cognates appear in the stimulus list.

2. Factors affecting cognate processing in bilinguals

The processing of cognates relative to non-cognate control words has been found to be influenced by a number of factors uncovered mostly with bilingual speakers. Due to their influence, cognate facilitation may decrease, it may become undetectable in the task, or the facilitation may even turn into inhibition (see Lijewska, 2020, for a review). Factors operating at the word level include cross-language overlap in cognates (Comesaña et al., 2015, 2018, 2021; Dijkstra et al., 2010; Schwartz et al., 2007; Schwartz & Tarin, 2021; Vanlangendonck et al., 2020), cognates' word class (Bultena et al., 2014; Van Assche et al., 2013), word frequency (Nakayama et al., 2013; Petrescu et al., 2017) or the type of script (Hoshino & Kroll, 2008). At the level of the task, the factors are task type (Bultena et al., 2014; Dijkstra et al., 2010; Muscalu & Smiley, 2019), stimulus list composition (Comesaña et al., 2015; Poort & Rodd, 2017; Vanlangendonck et al., 2020), the presence of the semantic context (Duyck et al., 2007; Lauro & Schwartz, 2017; Libben & Titone, 2009; Schwartz & Tarin, 2021), the language of the task (L1 vs. L2) (Cop et al., 2017; Palma et al., 2020) or task modality (Cornut et al., 2021; Frances et al., 2021). At the level of the speaker, cognate processing can be influenced by speaker's L2 proficiency/language dominance (Bultena et al., 2014; Nakayama et al., 2013) or type of their language background (Iniesta et al., 2021).

It needs to be noted that numerous studies (including many of the reviewed above) have tested both identical and non-identical cognates without distinguishing them in the analyses or the reported effects. However, those studies which did look into the role of the degree of cross-language overlap in processing, have shown that greater similarity generally tends to induce greater advantage, with the effect being discontinuously large for identical cognates (Dijkstra et al., 2010; Duyck et al., 2007). Still, non-identical cognates have been found to lead to either reduced facilitation (Dijkstra et al., 2010; Van Assche et al., 2011), null results (Duyck et al., 2007; Schwartz et al., 2007) or even inhibition depending on the variables such as the task at hand and stimulus list composition (Comesaña et al., 2012, 2015, 2018; Dijkstra et al., 2010; Poort & Rodd, 2017). Although even a relatively small orthographic form overlap across languages (as in *GUIDE*–*GIDS* and *RHYTHM*–*RITME*) has been found to yield facilitation in recognition by Dutch-English bilinguals (Dijkstra et al., 2010, p.298), previous research has still indicated that the level (and type) of form similarity across languages in non-identical cognates is a complex issue in processing. On the one hand, non-identical cognates which are highly similar in orthography and in phonology tend to quite consistently yield a processing advantage relative to non-cognates. On the other hand, when cognates are highly similar in phonology but to a smaller degree in orthography (or vice versa), they may either display effects in processing (Carrasco-Ortiz et al., 2021; Iniesta et al., 2021; Van Assche et al., 2011) or not (Carrasco-Ortiz et al., 2021; Schwartz et al., 2007).

A critical finding in the context of the current study was reported by Comesaña et al. (2015) with respect to the role of stimulus list composition in cognate processing exhibited by bilingual speakers. The authors selected identical cognates, non-identical cognates and non-cognate controls to orthogonally manipulate the level of cross-language overlap in orthography and phonology. Comesaña et al.'s identical cognates were identical across Catalan and Spanish either in orthography and in phonology (O+ P+; e.g., *PLATA*–*PLATA* ‘silver’) or identical in orthography and only highly similar in phonology (O+ P–; e.g., *JUGADOR*–*JUGADOR* ‘player’). In turn, the O+ P+ non-identical cognates (e.g., *BRUSA*–*BLUSA*, ‘blouse’) differed by one letter across Spanish and Catalan and had very similar pronunciations in the two languages, the O– P+ non-identical cognates (e.g., *TASCA*–*TAREA*, ‘task’) were still very similar phonologically but their cross-language overlap in orthography was lower, the O+ P– non-identical cognates (e.g., *ATZAR*–*AZAR*, ‘random’) had the opposite characteristics, and finally the cross-language overlap in the O– P– non-identical cognates (e.g., *CONTE*–*CUENTO*, ‘tale’) was lower both in phonology and in orthography.

The results of Experiment 1 (testing all types of cognates with matched non-cognate controls) revealed that only identical cognates exhibited cognate facili-

tation in the LDT performed in the less dominant language. The non-identical cognates yielded null effects. However, when identical cognates were removed from the stimulus list in Experiment 2, non-identical cognates yielded inhibition, i.e., the non-identical cognates were recognised more slowly than controls. Moreover, the authors also found that inhibition increased with growing mismatches between cognates' overlap in orthography and in phonology. Comesaña et al., (2012) found similar results in a study with unbalanced European Portuguese-English bilinguals (but see Schwartz et al., 2007, for an opposite pattern).

Comesaña et al. (2015) claimed that in a task testing only non-identical cognates relative to controls, cognates' orthographic and phonological representations compete for selection via lateral inhibition. This leads to slower responses to non-identical cognates relative to controls (i.e., inhibition in RTs). In contrast, when identical and non-identical cognates appear within a single stimulus list (as in Experiment 1), the non-target language gets highly activated (by the full form overlap in identical cognates); thereby the language context of the list induces null effects for non-identical cognates instead of inhibition. Thus, the inhibitory effects for non-identical cognates relative to controls observed in Experiment 2 were caused by "higher suppression of the non-target language when identical cognates [were] excluded from the list" (Comesaña et al., 2015, p. 622). The authors postulated that these effects could only be accounted for within BIA+ if it allowed for the top-down effect of language context, which so far has not been incorporated into the model. Therefore, the present study aimed to test whether inhibition of non-identical cognates reported for bilinguals by Comesaña et al.'s (2015) can be also obtained with trilingual participants. Trilinguals were particularly interesting because during word recognition they need to suppress not one (as bilinguals) but two non-target representations. Consequently, if Comesaña et al.'s (2015) proposition is correct, trilinguals should exhibit particularly robust inhibitory effects during recognition of non-identical cognates (especially the triple ones). Obtaining such robust inhibition with trilinguals would provide further evidence for the modifications within BIA+/Multilink postulated by Comesaña et al. (2015).

3. Factors affecting cognate processing in trilinguals

Studies investigating cognate processing by trilingual speakers are significantly less numerous than those testing bilinguals. Nevertheless, the available trilingual research has evidenced the influence of the following factors affecting cognate processing: the status of languages sharing a given cognate, i.e., cognates shared between the native language and a foreign language tend to yield different effects from the cognates shared between two foreign languages (Lijewska &

Błaszowska, 2021; Szubko-Sitarek, 2015; Van Hell & Dijkstra, 2002), the number of languages sharing a given cognate (double vs. triple cognates; Lemhöfer et al., 2004; Szubko-Sitarek, 2015), semantic context (Lijewska, 2023; Lijewska & Chmiel, 2015), the language of the task (Dijkstra & Van Hell, 2003; Poarch & Van Hell, 2012, 2014; Szubko-Sitarek, 2011), stimulus list composition (Zhu & Mok, 2020), participants' language background (Poarch & Van Hell, 2012, 2014) and proficiency (Lijewska & Błaszowska, 2021; Poarch & Van Hell, 2012, 2014; Szubko-Sitarek, 2015; Van Hell & Dijkstra, 2002).

Lemhöfer et al. (2004) reported a study with Dutch–English–German trilinguals who performed LDT in their L3 with two types of identical cognates and German control words. The stimuli included double L1–L3 cognates (overlapping across Dutch and German) and triple L1–L2–L3 cognates (overlapping across all the three languages). The results indicated that all cognates yielded shorter RTs and higher accuracy than controls but the advantage was more robust for triple cognates than for the double ones.

Similar findings were later reported by Szubko-Sitarek (2011, 2015) who conducted a series of LDTs with Polish learners of English as L2 and German as L3. She investigated cognate advantage induced by L1–L3 cognates, L2–L3 cognates and L1–L2–L3 cognates relative to controls in LDTs in L3 and in L1 (conducted with participants with a less or a more balanced proficiency in L2 and L3). All the triple cognates were identical whereas all types of double cognates included both identical and non-identical items (never analysed separately). Szubko-Sitarek found that triple cognates produced significantly shorter RTs and lower error rate when tested in L1 and in L3 but L1–L3 double cognates were recognised faster and more accurately than controls only in L3. The L2–L3 double cognates yielded no significant effects. Thus, the results demonstrated that cognate advantage depended on the type of cognate, the number of the languages sharing a given cognate, participants' proficiency in the non-native languages as well as the language of the task (for similar evidence in a translation task see (Lijewska & Błaszowska, 2021; Lijewska & Chmiel, 2015).

More recently, in an LDT in L2, Zhu and Mok (2020) found the typical pattern of cognate facilitation in RTs and in error data. When cognates and non-cognates were tested in the L3 LDT, Zhu and Mok found significantly lower accuracy for identical cognates relative to non-cognates (but no effects in RTs) and no effects whatsoever for non-identical cognates. They ascribed the lower accuracy to the difficulties experienced by less proficient trilinguals who likely hesitated deciding whether identical cognates are English (L2) or German (L3) words. Zhu and Mok subsequently tested their trilinguals in yet another L3 LDT where the stimulus list included L2–L3 cognates and L2–L3 interlingual homographs (words sharing form but not meaning across L2 and L3) relative to L3 con-

trol words. The results revealed significant facilitation for cognates but null effects for homographs relative to controls. The authors did not clarify whether this experiment tested identical and/or non-identical cognates. Zhu and Mok concluded that visual word recognition in trilinguals might be affected by an interplay of language dominance, stimulus list composition, orthographic similarity, performance strategy in a task, typological proximity between languages, and individual variations in language proficiency. The authors (2020, p.466) suggested that “future research is needed to tease apart these interactions in order to better understand trilingual non-native language processing”. Therefore, the present study makes the first step towards this goal by testing solely non-identical cognates in L3 LDTs.

4. The present study

The present study aimed to test processing of two types of double non-identical cognates (in Experiment 1) and triple non-identical cognates (in Experiment 2) relative to matched non-cognate control words in L3 LDTs with Polish–English–Dutch trilinguals. We selected the three types of cognates as they have been found to yield discrepant effects in LDTs performed by trilinguals (Szubko-Sitarek, 2015; Zhu & Mok, 2020). Double and triple cognates were tested in separate experiments in order not to amplify the influence of cross-language similarity (Comesaña et al., 2012). Importantly, both experiments included only non-identical cognates in the stimulus lists (Comesaña et al., 2012, 2015).

Formulating unequivocal predictions for the current study was far from trivial, given past evidence from research with non-identical cognates. In line with findings reported by e.g., Dijkstra et al. (2010) or Van Assche et al. (2011), one could expect a relatively small (due to incomplete form overlap) but still significant cognate facilitation across all the tested cognate types. Alternatively, in line with findings from studies involving only non-identical cognates (Comesaña et al., 2012, 2015), cognate inhibition, rather than facilitation, could be expected. If such inhibitory effects were indeed found with trilinguals, that would provide further evidence for the need of incorporating top-down effects of language context within BIA+/Multilink.

In Experiment 1, we expected equally robust cognate effects (facilitation or inhibition) for both types of double cognates. Even though the typological nature of Polish, English and Dutch vocabulary allowed us to select fewer L1–L3 cognates and with smaller overlap than the L2–L3 ones, we expected the influence of L1 (as the strongest of all participants’ languages), to be robust enough to surface as a significant cognate effect. In turn, in Experiment 2, triple non-identical cognates

were expected to yield especially pronounced effects because their recognition in L3 is affected by two other form-similar representations—one in L1 and one in L2.

4.1 Experiment 1

4.1.1 Participants

Forty-six (36 females, 10 males) Polish native speakers who learned English and Dutch as foreign languages took part in Experiment 1. They were all students of the Dutch Studies at Adam Mickiewicz University, Poznan, Poland. At the time of the experiment the participants were having classes mainly in Dutch and Polish, and one course in English. Questions adapted from the language history questionnaire (LHQ 2.0; Li et al., 2014) helped to discover participants’ language profile. Self-rating of language proficiency with respect to reading, writing, listening, and speaking used a 7-point Likert scale (1=very poor, 7= native-like). Four participants self-reported relatively high proficiency in other languages. Thus, in order to limit the possible influence of uncontrolled cross-language similarity, we decided to exclude data of those participants from further analyses. Other foreign languages that some of the remaining participants reported to know included: Spanish, German, Russian, French, Italian, Catalan, Danish, but as those languages

Table 1. Basic characteristics of the participants in Experiment 1 and 2 (mean values; SDs in parentheses)

Age	Experiment 1			Experiment 2		
	23.1 (2.1)			22.4 (1.6)		
Self-reported proficiency (scale 1–7)	English	Dutch	<i>p</i> -value (<i>t</i> -test)	English	Dutch	<i>p</i> -value (<i>t</i> -test)
Reading	5.9 (0.1)	5.6 (0.8)	.14	5.5 (1.4)	5.1 (1.3)	<.001
Writing	5.2 (0.1)	5.2 (0.8)	.31	4.7 (1.5)	4.5 (1.3)	.27
Speaking	5.3 (0.1)	4.9 (1.1)	.06	4.8 (1.6)	4.3 (1.3)	.02
Listening	5.7 (0.7)	5.0 (0.9)	<.01	5.3 (1.5)	4.4 (1.4)	<.001
Age of onset of language learning	8.2 (3.2)	19.0 (2.6)	<.001	7.3 (3.0)	18.6 (4.5)	<.001
LexTALE score	71.2 (12.0)	66.7 (7.4)	.03	70.7 (10.7)	63.5 (8.0)	<.001

were reported to be known at relatively low levels of proficiency and rarely used, thus they were not considered. The participants also completed the English and the Dutch versions of LexTALE (Lemhöfer & Broersma, 2012) and their LexTALE score was used as an independent measure of their proficiency in English and Dutch (but see Puig-Mayenco et al., 2023). Table 1 present participants' language characteristics. All but 3 participants were right-handed.

4.1.2 *Materials*

There were 42 double cognates, 42 non-cognates and 84 non-words which were used as stimuli in Experiment 1. Twenty-eight cognates, out of the 42 double cognates, overlapped across English and Dutch (L2–L3 cognates), and 14 cognates overlapped across Polish and Dutch (L1–L3 cognates). We specifically used only nouns in experimentation as they tend to yield most robust cognate effects (Bultena et al., 2014). The selection of cognates and non-cognates for Experiment 1 and Experiment 2 was a two-step process—an orthographic analysis and a norming study.

Orthographic analysis

From two databases (Otwinowska, 2015; Schepens et al., 2012), and two dictionaries (Klimaszewska et al., 2009; Martin et al., 1999), two independent judges selected an initial list of 208 nouns with formal similarity across Dutch and Polish, Dutch and English, and across Dutch, Polish and English. Next, each of the pre-selected similar items was matched in length and frequency with up to 3 nouns (dissimilar across Polish, English and Dutch). This resulted in a list of 342 items which were subsequently paired with their Polish and English translation equivalents and assessed in a norming study.

Norming study

The norming sessions were carried out either on-line (Instantly Inc., 2008) or in a pen-and-paper form. Word pairs were assessed in terms of formal similarity across English and Dutch and with respect to meaning equivalence by 25 Polish–English–Dutch speakers (none of them participated later in Experiment 1 or 2). Following Tokowicz et al. (2002), the participants were asked to complete 2 ratings—one to assess the similarity of word pairs with respect to both orthography and phonology (in one single rating) on a 7-point Likert scale (1–dissimilar; 7–very similar) and another one to assess semantic equivalence of the word pairs (1–dissimilar; 7–identical). Similar ratings were administered across Polish and Dutch with a new group of 20 trilinguals.

A summary of the characteristics of the selected targets used in Experiment 1 is presented in Table 2. Both types of the selected double cognates were matched with control words (non-cognates) with respect to the dominant part of speech,

frequency (in Zipf scale; van Heuven et al., 2014), length (in number of letters) and orthographic neighbourhood density in L3. The Polish–Dutch (L1–L3) double cognates (e.g., *SMAK–SMAAK*) and the English–Dutch (L2–L3) double cognates (e.g., *APPLE–APPEL*) used in Experiment 1 significantly differed from their matched non-cognate controls (e.g., *JASJE–JACKET*) with respect to the rated form similarity. L1–L3 cognates and their matched controls were marginally less frequent in L3 but significantly longer, and had fewer orthographic neighbours than the L2–L3 cognates and their matched controls (see Table 2). A complete list of cognates and non-cognates used in Experiment 1 can be found in OSF (Lijewska & de Louw, 2024).

Table 2. A summary of stimulus characteristics (mean values; SD in parentheses) as well as independent-samples *t*-test *p*-values for comparisons between stimulus types

Stimulus type	L1–L3			L2–L3			L1–L3 vs. L2–L3 <i>p</i> -value (<i>t</i> -test)	
	Cognates	Non-cognates	<i>p</i> -value (<i>t</i> -test)	Cognates	Non-cognates	<i>p</i> -value (<i>t</i> -test)	Cognates	Non-cognates
Length (in no. of letters)	7.4 (1.7)	7.4 (1.7)	1.00	5.1 (1.0)	5.1 (1.0)	1.00	<.001	<.001
Frequency (in Zipf-values)	3.9 (0.7)	4.0 (0.7)	.69	4.3 (0.6)	4.3 (0.5)	.70	.05	.15
Rated similarity	4.7 (1.0)	1.4 (0.5)	<.001	5.9 (0.6)	1.7 (0.6)	<.001	<.001	.06
OND	2.0 (3.9)	1.4 (3.0)	.28	6.0 (4.8)	4.7 (3.6)	.67	.02	<.01
nLD	0.4 (0.2)	0.8 (0.2)	<.001	0.3 (0.1)	0.9 (0.1)	<.001	.06	.38

Note. OND=orthographic neighbourhood density, nLD=normalised Levenstein distance. Stimulus selection was performed only on the basis of cross-language similarity ratings. Since normalised nLD scores (Schepens et al., 2012) are a commonly used measure of cross-language similarity in cognates (Costa et al., 2022), the nLD values for the selected stimuli have been added to Table 2 only for ease of comparison with other studies.

We additionally checked the selected items for possible influences of uncontrolled word characteristics by retrieving RTs for the stimulus words from the database of the Dutch Lexicon Project 2 (Brysbaert et al., 2016). The statistical analysis of the RTs (by way of *t*-tests) revealed no significant differences between all cognates and their matched controls (all *ps*>.05). Non-words were created with the help of Wuggy (a pseudoword generator; Keuleers & Brysbaert, 2010). The selected non-words matched the cognates and non-cognates in length.

4.1.3 Procedure

The participants read instructions (in Dutch) displayed on the computer screen. Instructions were next repeated orally in Polish (participants' L1) by the experimenter in order to make sure that all the participants understood them correctly. The participants were instructed to decide (by pressing buttons) whether a given letter string displayed on the computer screen was a Dutch word or not. The participants were asked to respond as quickly and as accurately as possible. The "yes" response was given with the right hand. The presentation of the stimuli within the training session and during the experiment proper was randomised for all participants.

Before the experiment proper started, the participants performed a training session with 20 items (not used in the experiment proper). If the required 75% accuracy level was achieved during the training session, the experiment proper followed. Otherwise, the participant had to repeat the training session.

Within each trial first, the fixation point appeared in the centre of the screen for 500 ms. Then, a target word was displayed in the same location and remained on the screen until the participant pressed one of the buttons or until 3000 ms passed. The intertrial interval was 500 ms.

After the main experiment, the participants filled in the LHQ questionnaires and performed LexTALE tests in English and Dutch (Lemhöfer & Broersma, 2012). All participants signed informed-consent forms before the study and received course credit or chocolate bars as a thank-you gift for their participation.

4.1.4 Data selection and analysis

The analysis of reaction times (RTs) was performed for accurate trials only (4.05% of observations eliminated as inaccurate from the analysis of RTs). Trials where RTs exceeded $2.5 \times \text{SD}$ around the participant and condition mean were considered outliers and eliminated from further analyses (3.25% of the data). Due to the typical skewing of processing times toward longer durations, BoxCox analysis was performed, resulting in inverse transformations ($1/\text{DV}$) applied to the RTs. The resulting frequency distributions were close to normal. In order to avoid very small values and to obtain coefficients with the same sign as in models fitted to untransformed data, the inversely-transformed RTs were multiplied by -1000 . The accuracy data were analysed using a logit-link function, as recommended by Jaeger (2008). The RTs were analysed using linear mixed-effects models from the lme4 library (Bates, Mächler, et al., 2015) of the R system for statistical computing (R Core Team, 2020). Significance values for fixed effects were calculated using the lmerTest package of R (Kuznetsova et al., 2015).

Since L1–L3 cognates and their controls significantly differed from L2–L3 cognates and controls across a number of word characteristics, the two types of cognates and controls were analysed in separate models (always with the fixed factor Cognateness¹ having two levels coded as 1=non-cognates vs. 2=cognates). All models used successive difference contrasts (contr.sdif) where the intercept represents the grand mean (Schad et al., 2020).

At the outset, all statistical models included a maximal random factor structure for subjects: the intercepts and slopes for the factor Cognateness and their correlations as well as random intercepts for items. The models included also word frequency and word length (in Polish, English and Dutch) as mean-centred covariates. Since the participants differed in their L2 and L3 proficiency levels as indicated by LexTALE scores, the scores were also added as mean-centred covariates. The random factor principal component analysis (with the rePCA function), as suggested by Matuschek et al. (2017), was used to check against overparameterization of the models. Next, the resulting model was compared to the minimal model (only with random intercepts). The comparison of goodness-of-fit of the models (as indicated by log-likelihood ratio) showed that the parsimonious model (Bates, Kliegl, et al., 2015) included random intercepts and slopes for Cognateness for subjects and random intercepts for items. The covariates were excluded from the model also on the basis of the comparisons of goodness-of-fit of models with and without covariates (as indicated by the log-likelihood ratio). The final models are reported in Table 4 and 5. The examination of the frequency distribution of random factor values (with the qqPlot function of library car, Fox & Weisberg 2010) revealed near-normally distributed values for the analysis. The selection of the most parsimonious mixed-effects logistic regression model for accuracy data was performed in a similar fashion. The final models (with random intercepts for participants and items) for accuracy data are reported in Table 6 and 7. The R script of the analysis is in OSF (Lijewska & de Louw, 2024). For the sake of clarity of presentation, data tables and figures are based on untransformed (observed) data.

4.1.5 Results

Table 3 presents observed mean RTs values (with SDs) and error rates recorded in Experiment 1. The analysis of RTs revealed a significant cognate facilitation effect where L2–L3 cognates were recognised 42 ms faster than matched non-cognates. Even though the facilitation in RTs for L1–L3 cognates vs. matched controls failed

1. All data from Experiment 1 and 2 were additionally analyzed in models with a continuous predictor *Rated cross-language similarity* (instead of the categorical factor Cognateness), but no new effects were revealed.

to reach significance, the numerical values still trended more towards facilitation than inhibition (see Table 4 and 5). The analysis of the accuracy data revealed no significant effects of cognates (see Table 6 and 7).

Table 3. Mean RTs; (SDs in parentheses) as well as error rates (ER, as % error) for L1–L3 as well as L2–L3 cognates and matched non-cognates

Target type	L1–L3		L2–L3	
	RT	ER	RT	ER
cognates	843 (346)	10.0	694 (199)	2.3
non-cognates	875 (340)	1.9	736 (225)	4.3
effect	–32	8.1	–42	–2.0

Note. The significant effect has been bolded.

Table 4. Summary of the linear mixed-effects model for RTs for L1–L3 cognates and controls

<i>Predictors</i>	–1000 * invRT			
	<i>b</i>	<i>SE</i>	<i>t</i>	<i>p</i>
(Intercept)	–1.30	0.04	–31.11	<.001
Cognateness	–0.05	0.04	–1.02	.309
Frequency in L3 (centred)	–0.15	0.04	–3.89	<.001
Length in L3 (centred)	0.04	0.01	2.95	.003
Random Effects				
σ^2	0.07			
τ_{00} Subject	0.05			
τ_{00} Target	0.01			
τ_{11} Subject.Category2–1	0.01			
ρ_{01} Subject	–0.40			
ICC	0.49			
N_{Subject}	42			
N_{Target}	28			
Observations	1075			
Marginal R^2 / Conditional R^2	.144 / .567			

Note. The *p*-values for significant predictors have been bolded.

Table 5. The summary of the linear mixed-effects model for RTs for L2–L3 cognates and controls

<i>Predictors</i>	–1000 * invRT			
	<i>b</i>	<i>SE</i>	<i>t</i>	<i>p</i>
(Intercept)	–1.49	0.04	–39.60	<.001
Cognateness	–0.07	0.03	–2.55	.011
Frequency in L2 (centred)	–0.10	0.04	–2.51	.012
Length in L2 (centred)	–0.01	0.01	–0.96	.336
Frequency in L3 (centred)	–0.02	0.03	–0.44	.657
Length in L3 (centred)	0.04	0.01	2.55	.011
Random Effects				
σ^2	0.07			
τ_{00} Target	0.01			
τ_{00} Subject	0.05			
τ_{11} Subject.Cognateness2–1	0.00			
ρ_{01} Subject	0.42			
ICC	0.46			
N_{Subject}	42			
N_{Target}	56			
Observations	2194			
Marginal R^2 / Conditional R^2	.041 / .483			

Note. The *p*-values for significant predictors have been bolded.

Table 6. The summary of the mixed-effects logistic regression model for accuracy data for L1–L3 cognates and controls

<i>Predictors</i>	Accuracy			
	<i>Odds Ratios</i>	<i>SE</i>	<i>z</i>	<i>p</i>
(Intercept)	142.86	82.37	8.61	<.001
Cognateness	0.21	0.24	–1.37	.172
Random Effects				
σ^2	3.29			
τ_{00} Subject	0.36			
τ_{00} Target	5.53			
ICC	0.64			
N_{Subject}	42			
N_{Target}	28			
Observations	1176			
Marginal R^2 / Conditional R^2	.062 / .664			

Note. The *p*-values for significant predictors have been bolded.

Table 7. The summary of the mixed-effects logistic regression model for accuracy data for L2–L3 cognates and controls

<i>Predictors</i>	<i>Accuracy</i>			
	<i>Odds Ratios</i>	<i>SE</i>	<i>z</i>	<i>p</i>
(Intercept)	108.42	33.10	15.35	<.001
Cognateness	0.64	0.35	−0.82	.413
<i>Random Effects</i>				
σ^2		3.29		
τ_{00} Target		2.30		
τ_{00} Subject		0.70		
ICC		0.48		
N_{Subject}		42		
N_{Target}		56		
Observations		2352		
Marginal R^2 / Conditional R^2		.008 / .481		

Note. The *p*-values for significant predictors have been bolded.

In order to make the pattern of results obtained in Experiment 1 clearer, we decided to first report findings from Experiment 2. All results will then be collectively discussed in the General Discussion. The goal of Experiment 2 was to test L1–L2–L3 non-identical cognates with a group of Polish–English–Dutch trilinguals.

4.2 Experiment 2

4.2.1 Participants

Forty-one (31 female, 10 male) Polish learners of English and Dutch as foreign languages participated in Experiment 2. They were recruited from the same participant pool as the ones who took part in Experiment 1 and they provided the same language background information. The statistical comparison of participants’ characteristics in both experiments revealed significant differences only in self-reported L3 proficiency levels ($ps < .05$), which was however not confirmed in the analysis of independent measure of L3 proficiency – LexTALE scores ($p > .05$). Despite screening, once data collection was over, it turned out that 7 participants of Experiment 2 had taken part in Experiment 1 completed a month earlier. Since in terms of targets, the two experiments overlapped only in 6 non-cognates, we decided not to exclude data from the seven participants but we did exclude those participants’ data for the overlapping items. The participants’ characteristics are summarised in Table 1.

4.2.2 Materials

Forty-three triple (L1–L2–L3) non-identical cognate nouns (e.g., *GRUPA–GROUP–GROEP*) and 43 non-cognate nouns (e.g., *BROEK–TROUSERS–SPODNIE*) matched in length, frequency (Zipf scale), orthographic neighbourhood density in Dutch and concreteness were selected on the basis of the norming study reported in Section 4.1.2. The cognates significantly differed from their matched non-cognate controls with respect to the rated form similarity and orthographic similarity measured by the normalised Levenstein distance (nLD; Schepens et al., 2012; all $ps < .05$). In turn, the comparison of nLD scores for the triple cognates themselves showed that their nLD scores across Polish and Dutch were not significantly different than across English and Dutch (all $ps > .05$). However, the comparison of the triple cognates’ form similarity ratings (combining the assessment of orthographic and phonological similarity) showed that the cognates were rated as significantly less similar across Polish and Dutch than across English and Dutch. Additionally, 86 non-words (matching the words in length) were selected following the same procedure as described in Experiment 1. Table 8 summarizes the characteristics of the stimuli used in Experiment 2. A complete list of cognates and non-cognates used in Experiment 2 can be found in OSF (Lijewska & de Louw, 2024).

Table 8. A summary of the characteristics of the stimuli used in Experiment 2 (mean values; SD in parentheses) as well as independent-samples t -test p -values for comparisons between the stimuli

Characteristics	L1–L2–L3 cognates	Non-cognates	p -values
Length (in Zipf values)	6.2 (1.3)	6.2 (1.3)	.93
Frequency (in letters)	4.1 (0.5)	4.1 (0.4)	.82
OND	2.4 (3.2)	3.0 (4.3)	.49
Rated form similarity across Polish and Dutch	4.8 (0.8)	1.1 (0.2)	<.001
Rated form similarity across English and Dutch	5.3 (0.6)	1.2 (0.3)	<.001
nLD ^a across Polish and Dutch	0.4 (0.2)	0.9 (0.1)	<.001
nLD ^a across English and Dutch	0.3 (0.1)	0.9 (0.1)	<.001

Note. OND=orthographic neighbourhood density, nLD=normalised Levenstein distance
a. Similarly to Experiment 1, nLD scores were not used for stimulus selection but have been added in Table 8 for ease of comparison of the current stimuli with other studies.

4.2.3 Procedure

The procedure was identical to the one employed in Experiment 1.

4.2.4 Data selection and analysis

After the initial analysis of experimental data with respect to accuracy, 1 further participant had to be eliminated from further analyses as they failed to reach 75% accuracy level in the experiment proper. The RTs and accuracy data in Experiment 2 were analysed using the same software, packages and procedures as in Experiment 1. Of note, 3.06% of RTs in Experiment 2 were excluded as inaccurate responses and further 3.5% were excluded as outliers. The random structure of the most parsimonious linear mixed-effect model for RTs and the mixed-effect logistic regression for accuracy included only intercepts for subjects and items. The R script of the analyses is listed in OSF (Lijewska & de Louw, 2024).

4.2.5 Results

Table 9 presents observed mean RTs values (with SDs) and error rates recorded in Experiment 2. The analysis of RTs (see Table 10) revealed that cognates were recognised significantly faster than non-cognates. As far as accuracy data is concerned, no significant cognate effects were found (see Table 11).

Table 9. Mean RTs; (SDs in parentheses) as well as error rates (ER, as % error) for triple cognates and matched non-cognates

Target type	RT	ER
cognates	728 (209)	3.0
non-cognates	772 (225)	3.1
effect	-44	-0.1

Note. The significant effect has been bolded.

Table 10. Summary of the linear mixed-effects model for RTs collected in Experiment 2

<i>Predictors</i>	−1 000 * invRT			
	<i>b</i>	<i>SE</i>	<i>t</i>	<i>p</i>
(Intercept)	−1.42	0.03	−43.03	<.001
Cognateness	−0.08	0.02	−3.83	<.001
Frequency in L3 (centred)	−0.14	0.02	−6.04	<.001
Random Effects				
σ^2	0.07			
τ_{00} Target	0.01			
τ_{00} Subject	0.04			
ICC	0.42			
N _{Subject}	40			
N _{Target}	86			
Observations	3 182			
Marginal R ² / Conditional R ²	.048 / .446			

Note. The p-values for significant predictors have been bolded.

Table 11. The summary of the mixed-effects logistic regression model for accuracy data collected in Experiment 2

<i>Predictors</i>	Accuracy			
	<i>Odds Ratios</i>	<i>SE</i>	<i>z</i>	<i>p</i>
(Intercept)	177.56	53.65	17.14	<.001
Cognateness	1.32	0.67	0.56	.577
Frequency in L3 (centred)	10.40	6.04	4.03	<.001
Length in L3 (centred)	1.68	0.33	2.68	.007
Random Effects				
σ^2	3.29			
τ_{00} Target	2.39			
τ_{00} Subject	0.51			
ICC	0.47			
N _{Subject}	40			
N _{Target}	86			
Observations	3 398			
Marginal R ² / Conditional R ²	.178 / .563			

Note. The p-values for significant predictors have been bolded.

5. General discussion

The aim of the present study was to test double and triple non-identical cognates in L3 LDTs without any identical cognates in the stimulus lists. A significant cognate advantage was found in RTs for double L2–L3 non-identical cognates and for triple non-identical cognates (L1–L2–L3) relative to matched control words. Even though L1–L3 cognates failed to induce any significant effects, they trended (numerically) towards a cognate advantage in RTs. Thereby we found evidence against the contention put forward by Comesaña et al. that non-identical cognates are subject to lateral inhibition during recognition when identical cognates are excluded from the task. Actually, the recorded significant facilitation in RTs for L2–L3 cognates and for the L1–L2–L3 ones relative to matched controls suggests that, in line with BIA+ and Multilink (Dijkstra et al., 2019; Dijkstra & van Heuven, 2002), similar form representations across the languages got co-activated and jointly activated the semantic representations thereby leading to speeded responses to cognates relative to non-cognate control words.

Although the lack of statistically significant effects for L1–L3 cognates found in our study could be claimed to stem from lateral inhibition as proposed by Comesaña et al. (2015), there are other explanations that appear to be more likely for our data, especially in view of the facilitatory trend in the RTs for L1–L3 cognates. Due to cross-language differences between Polish and Dutch, the L1–L3 (Polish–Dutch) cognates employed in Experiment 1 were less numerous and their assessed cross-language similarity was significantly lower than was the case for L2–L3 cognates. Therefore, the relatively low cross-language similarity level for L1–L3 cognates (Dijkstra et al., 2010) and power problems could have led to the observed non-significant effects (Brysbaert, 2019). In additional post-hoc analyses we selected an equal number (12) of L1–L3 and L2–L3 cognate+control pairs with similar levels of the rated cross-language similarity. We applied the Bonferroni correction due to multiple tests with the same data (von der Malsburg & Angele, 2017), which set the alpha level at $p < .025$. The L1–L3 and L2–L3 cognate advantage was not significant ($p = .027$), showing that the null effects for L1–L3 cognates could have indeed stemmed from power problems in the data.

Furthermore, the interpretation that the obtained null effect for L1–L3 is due to lateral inhibition in recognition is difficult to defend in view of the fact that the stimulus list in our experiments contained no identical cognates. It is precisely this type of language context for which Comesaña et al. (2015) posulated increased lateral inhibition surfacing as significantly slower recognition of non-identical cognates relative to controls (i.e., significant inhibition). Moreover, since our participants completed the LDT in their least dominant language (L3), the L1 (as the most dominant language) should have been particularly

strongly suppressed (Blanco-Elorrieta & Pylkkänen, 2018; Gollan & Ferreira, 2009; Green, 1998) thus making the inhibitory effects of L1–L3 cognates particularly pronounced in RTs. Since this was not the case in our data, it is unlikely that the null results recorded for our L1–L3 cognates were caused by lateral inhibition. Moreover, if lateral inhibition is used to account for the inhibitory trend in accuracy, it cannot be reconciled with the facilitatory trend in RTs. If L1 was indeed particularly strongly suppressed in the task as postulated by Comesaña et al. (2015), there would be no reason to expect opposite trends in RTs and in accuracy.

Yet, the question remains why Comesaña et al. (2015) found significant inhibition while we obtained null results (with L1–L3 cognates) or significant facilitation (with L2–L3 cognates) when the tasks included only non-identical cognates. It might be that differences in the cross-language similarity levels between the L2–L3 cognates used in our study and the L1–L2 cognates used by Comesaña et al. (2015) underlie the discrepancy between the obtained effects. Our stimuli were normed with respect to the combined overlap in orthography and in phonology, whereas the similarity of the stimuli used by Comesaña et al. (2015) was analysed separately in terms of orthography and in terms of phonology. Thus, the differences in the adopted strategies for stimulus selection might have translated into differences in cross-language similarity levels across the selected stimulus sets, which, in turn, led to discrepant results.

The problem of the interplay of cognate similarity levels in orthography and phonology and their joint influence on cognate processing is far from clear. As has been mentioned in the Introduction, lower orthographic and phonological similarity in cognates may translate into smaller cognate facilitation (Dijkstra et al., 2010; Van Assche et al., 2011), null results (Duyck et al., 2007) or inhibition (Comesaña et al., 2012, 2015; Schwartz et al., 2007). Some studies show that the activation of phonology in cognates occurs only with sufficiently high overlap in orthography across languages (Inieta et al., 2021; Schwartz et al., 2007) and this may lead to inhibitory effects. In contrast, other experiments indicate that it is the degree of cross-language mismatch in orthography relative to phonology that induces most pronounced inhibitory effects (Comesaña et al., 2012, 2015). Actually, the comparison of the stimulus lists utilised across the studies suggests that a vast majority of our cognates most closely matched the O– P– cognates (see Introduction for details) employed by Comesaña et al. (2015). Since the role of the degree of the cross-language overlap in orthography/phonology is crucial for cognate facilitation/inhibition patterns (Comesaña et al., 2015; Dijkstra et al., 2010; Schwartz et al., 2007), the divergent degrees of stimulus similarity across the studies might have contributed to the discrepancy between the results.

However, we propose that there is yet another likely explanation for the observed discrepancy between our results and those of Comesaña et al. (2015). This explanation accentuates even more the key role of stimulus list composition in non-identical cognate effects. In our study, during stimulus selection a combined assessment of phonology and orthography was considered since the relation of orthographic and phonological similarity was not manipulated directly. This difference might have affected the participants' performance in that they focused more on the nuanced differences in the levels of cross-language similarity in orthography and phonology when taking the lexicality decisions (i.e., distinguishing between words and nonwords) in Comesaña et al.'s LDT than in ours. This, in turn, might have induced slower responses to the cross-linguistically similar cognates than to the non-cognates in Comesaña et al.'s data. However, since the authors reported no analyses of whether and how the inhibitory effects they found unfolded during their experiment, our interpretation remains only a proposition.

Alternatively, the discrepancy between the current data and the results reported by Comesaña et al. might be ascribed to differences in the language profiles of the tested participants (trilingual vs. bilingual) or even to differences in the language proximity between the languages tested across the studies. Even though such accounts cannot be ruled out, prior research has shown that bilinguals tend to show similar double cognate effects to trilinguals (Lijewska & Błaszowska, 2021; Poarch & Van Hell, 2012); also in tasks with typologically closer or more distant languages. Lijewska and Błaszowska (2021) recently reported a study where they asked Polish–German–English late trilinguals and German–English late bilinguals to translate non-identical Polish–English cognates, German–English cognates and matched controls from trilingual participants' L3 into their L1 and into their L2 and from bilingual participants' L2 into their L1. Both groups displayed a cognate advantage. For trilinguals the cognate advantage emerged only in English–Polish translation accuracy data and for bilinguals the advantage appeared only in English–German translation latencies. Since trilinguals' proficiency in German and English was significantly lower than the bilinguals' proficiency in the languages, the difference in the measures exhibiting the cognate advantage was assigned to the role of proficiency. Importantly, significant facilitation was identified across typologically closer languages–English and German in bilinguals as well as across more distant languages–Polish and English in trilinguals.

Furthermore, Comesaña et al. obtained similar inhibitory effects with early balanced bilinguals highly proficient in relatively similar languages (Spanish and Catalan; Comesaña et al., 2015) as well as with unbalanced sequential bilinguals with less similar languages (European Portuguese and English; Comesaña et al.,

2012). Actually, our trilingual participants seem more similar to the bilinguals tested by Comesaña et al. (2012) in that their onset of L2 acquisition was similar and the context of L2/L3 use also appeared to be comparable, but still the recorded cognate patterns differed significantly across the studies. Consequently, the explanation of the discrepancy between the effects identified in Comesaña et al. experiments and in ours likely resides in the nature of the stimuli rather than in the characteristics of the participants or the proximity of the tested languages.

Even though the present results do not match the effects reported by Comesaña et al.'s (2015), the significant facilitation we found in our experiments did indicate that multiple languages are co-activated when trilinguals perform LDT in their L3. This finding is generally in line with previous evidence reported for trilingual participants (Lemhöfer et al., 2004; Szubko-Sitarek, 2011, 2015; Van Hell & Dijkstra, 2002; Zhu & Mok, 2020). However, a closer scrutiny of the patterns of cognate effects reported in our study as compared to previous trilingual research reveals the specificity of the representation and processing of various types of cognates in trilinguals.

In the present study, in Experiment 1, our Polish–English–Dutch participants exhibited significant facilitation in their RTs for L2–L3 cognates but not for L1–L3 ones. Szubko-Sitarek (2015) found just the opposite pattern for her Polish–English–German participants. The discrepancy of the two sets of results most likely stems from differences in the designs across the experiments. Szubko-Sitarek combined identical and non-identical cognates (altogether 12 L1–L3 cognates and 12 L2–L3 ones and 24 matched non-cognates) in her L3 LDT performed by 17 trilinguals. Only 5 of her L1–L3 cognates and only 2 of L2–L3 ones were orthographically non-identical. We tested more cognates and controls with more participants but we used non-identical cognates only. The juxtaposition of our design and results with the Szubko-Sitarek's ones shows that significant facilitative effects emerge for L1–L3 cognates even with relatively few items and participants but once identical cognates are included in an L3 LDT. In contrast, significant effects can be obtained with L2–L3 cognates even when no identical items appear in the task but more items and participants need to be tested. A similar conclusion can be drawn when the current results for double non-identical L2–L3 cognates are compared to the findings from a more recent study by Zhu and Mok (2020). Since our dataset for non-identical cognates was larger than the ones analysed by Zhu and Mok, it is possible that more numerous observations are needed for the significant L2–L3 cognate effects to become detectable.

Furthermore, our participants were relatively proficient in L2 and in L3, which might have also contributed to the strength of the observed advantage. The influence of proficiency on the exhibited patterns of cognate facilitation has

already been attested in earlier trilingual research (Lijewska & Błaszowska, 2021; Poarch & Van Hell, 2014; Van Hell & Dijkstra, 2002).

As far as triple cognates are concerned, similarly to Szubko-Sitarek (2015) and Lemhöfer et al.'s (2004), we found significant cognate facilitation for our L1–L2–L3 cognates relative to matched controls. Even though a direct comparison of the effect sizes cannot be performed across the studies as neither of the other authors reported any direct measures of effect size, a comparison of the numerical values shows that our effects were considerably smaller than those reported earlier. As expected, non-identical triple cognates tested with trilinguals seem to yield more subtle facilitative effects than the identical ones. Similar observations have also been reported in bilingual studies (Dijkstra et al., 2010; Van Assche et al., 2011).

Even though the design of the current research did not include a direct test of triple and double cognates within a single task, it might be interesting to note that the recorded advantage for our double L2–L3 cognates was numerically very similar to the facilitation induced by our triple cognates. Consequently, it may seem attractive to attribute the triple cognate advantage solely to the influence of similarity across L2 and L3 within the triple cognates. In order to investigate this matter more closely, we compared (post-hoc) the similarity levels in cognates across the languages with a series of t-tests. The post-hoc analysis of nLD scores across languages revealed that our triple cognates were not significantly orthographically different from L1–L3 double cognates or from our double L2–L3 cognates (all $ps > .05$). In contrast, the triple cognates were rated as significantly less similar than the L2–L3 cognates ($p < .001$) (but not significantly different from the L1–L3 ones). As a consequence, the numerically similar 44-ms advantage obtained for triple cognates relative to the 42-ms advantage recorded for double L2–L3 cognates) likely stems from a complex interplay of similarity levels across all languages rather than from the similarity shared solely across L2 and L3. Nevertheless, a careful manipulation of cross-language similarity levels in triple and double non-identical cognates tested in a single task is needed before any reliable conclusions can be drawn with respect to the influence of cross-language similarity levels within triple non-identical cognates.

Finally, no significant cognate effects were recorded in accuracy data across all types of cognates employed in our study. Comesaña et al. (2015) also found null results in accuracy data when they tested non-identical cognates an L2 LDT with bilingual participants, even when no identical cognates appeared in the task. Our results for double cognates are also in line with prior research reported by Zhu and Mok (2020) for non-identical L2–L3 cognates, when these were tested alongside identical ones in an L3 LDT. However, our accuracy data for triple cognates failed to agree with Szubko-Sitarek's (2011) and Lemhöfer et al.'s (2004) results.

They both reported higher accuracy for their triple cognates than for controls. The discrepancy between their results and ours most likely resides in the type of items used. Unlike in the present study, Szubko-Sitarek as well as Lemhöfer et al. used only identical triple cognates in their research. Hence, it seems that non-identical ones, unlike identical ones, do not exhibit significant advantage in accuracy in LDTs, which lends further support for the claim that these two types of cognates might be represented differently in the mind, as proposed within BIA+ and Multilink (Dijkstra et al., 2010, 2019; Dijkstra & van Heuven, 2002).

6. Conclusions

The present study investigated the processing patterns of double and triple non-identical cognates tested with trilingual speakers of Polish as L1, English as L2 and Dutch as L3 in two L3 LDTs. Importantly, the tasks did not include any identical cognates in the stimulus lists. L2–L3 double cognates and L1–L2–L3 triple cognates yielded facilitation (only in RTs). These findings lend no support for the increased lateral inhibition of non-target language representations during recognition postulated earlier for bilinguals and proposed for implementation within BIA+ (Comesaña et al., 2015). On the contrary, the present results seem to support the current assumptions of BIA+ (and Multilink) concerning the mechanism of cognate facilitation. Furthermore, the present trilingual data emphasize even further the key role of the composition of the stimulus list in LDTs exploring the cognate effects. This composition might influence the observed effects not only in terms of the inclusion of identical vs. non-identical cognates in the task but also in terms of the interplay of various levels of cross-linguistic similarity within the tested cognates themselves.

In sum, the present study showed that non-identical cognates can on their own induce facilitation during word recognition in L3. However, stimulus list composition and varying levels of cross-language similarity in non-identical cognates may to a large extent affect the observed effects. These findings can be accommodated by the current versions BIA+ or Multilink, so no new mechanism are needed to model word recognition even in trilinguals.

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Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, or publication of this article.

Data availability statement

The participants of this study had not been asked to give written consent for their data to be shared publicly so the supporting data cannot be made available. In turn, the R code supporting the data analysis and the list of stimuli used in the study are available in OSF at https://osf.io/3tuda/?view_only=89bbc4c44b6a472d976odcecaa9f22c9.

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